

Scattered Spring: How Temperature Synchronizes Ecosystems and its Consequences for Climate Change

Jonathan Auerbach, Andrew Gelman, and E. M. Wolkovich

March 17, 2025

Abstract

A wide variety of biological processes are triggered by cumulative temperatures. Many plant events, such as leafout and flowering, occur once the temperature sum exceeds a threshold. In this way, cumulative temperatures may coordinate the timing of ecological events within a species and potentially ecosystem. But increasing reports of slowing responses to warming, increasing variance in the timing of spring events and other anomalies have raised questions about how reliable this temperature-based coordination is with anthropogenic climate change. Researchers have proposed a suite of varying hypotheses to explain these findings, all of which add complexity beyond a simple model of events triggered by cumulative temperatures. Yet this research has not fully evaluated predictions from the simple model.

Here, we address this gap by using renewal theory to model temperature-triggered biological processes as stopped random walks. Cumulative temperature determines the position of the walk, and the occurrence of the biological event is the stopping time. This framework yields two theory-driven implications for how warming alters the timing and synchrony of seasonal events. First, the relationship between temperature and the expected event time is nonlinear. As temperatures increase, the average springtime event such as flowering occurs earlier, but the rate at which they advance can diminish or even accelerate. Second, warming affects not only the expected time but its variability. As temperatures increase and springtime events shift from the equinox toward the solstice, springtime events become less coordinated, producing a more variable or “scattered” spring. We verify both implications empirically using experimental and observational data.

Introduction

Many biological events are triggered not by the temperature on a single day, but by the temperature accumulated over many days. A canonical example is the law of the flowering plants, which states that a bud blooms in the spring once cumulative temperatures exceed a plant-specific threshold. This and similar mechanisms synchronize a wide variety of seasonal events in ecosystems—for instance, ensuring that pollinators emerge when flowers are in bloom. They also support an array of forecasting tools in which scientists track cumulative temperature or “growing degree days” to predict harvest dates, manage pest

emergence, and assess how shifting temperature patterns reshape ecological communities with climate change. Shifts to earlier biological events—earlier phenology—has been the clearest ‘fingerprint’ of anthropogenic climate change across both natural and managed systems, but have recently become less predictable. In recent decades phenological advances have decelerated, advancing less with each degree of warming (fu2015declining). With this, variance in the timing of phenological events has appeared to both decline or increase, without a clear pattern (vitasse2018global,stemkovski2023disorder). At the same time, previously unknown or rare phenological events appear more common (chuine2025living). While research has provided myriad explanations for these events—often with directly contrasting predictions and logic (fu2015declining,wolkovich2021simple,gao2024thermal)—no work has examined predictions from the existing cumulative sum model fully.

Scientists have used a basic cumulative sum model for over two centuries, but have rarely if ever integrated the relevant mathematical tools developed in probability and statistics over past decades relevant to this model—specifically renewal theory designed for analyzing stopped random walks. Despite the shared mathematical structure, few have connected growing degree-day models and the modern limit theory of stopped random walks.

Here we draw on renewal theory to predict how events triggered by a cumulative temperature sum will shift with warming. Our model predicts observed advances as well as shifts in variance. We show predictions depend critically on characteristics of the underlying temperature regime—variance in the timing of biological events can decrease or increase with warming depending on whether temperatures during the period of accumulation are stationary (as in the winter and summer) or increasing (as in the spring). Taken together our results show how climate change can (i) move spring events earlier at an increasing or decelerating rate, and (ii) increase the variance of their timings, producing less synchronized events as they arrive earlier.

Results & Discussion

Spring events, such as plant flowering, occur on the date on which the cumulative sum first crosses a plant-specific temperature threshold. This places biological timing problems within the domain of stochastic process theory, allowing us to asymptotically characterize the distribution of event timings (hitting times) and thus predict temperature-triggered biological events.

We model effective temperatures (X_i)—the temperatures experienced by a plant bud at a given location—as a baseline temperature (α) with a within-season warming rate ($\{\epsilon_i\}_{i \geq 1}$) commonly seen in the spring months (Fig 1) and some noise (ϵ_i , thus $X_i = \alpha + \beta i + \epsilon_i$). Noise in this formulation could represent many biological aspects depending on the scale and our developing understanding of what triggers spring plant events. For example, in our focus on the bud-level it represents idiosyncratic variation at that bud, which could be driven potentially by microclimate variation (schwartz2014separating), including bud color variation

(peaucelle2022accurate), or internal differences that vary when the bud starts accumulating temperatures (CITESmoleculararchillpapers). Bloom occurs on the day (n) the cumulative temperature ($Z_n = \sum_{i=1}^n X_i$) where $t > 0$ is a plant-specific temperature-sum threshold; $\nu(t) = \min\{m : Z_m > t\}$. Thus $\nu(t)$ is the bloom date—and the hitting time at which cumulative temperatures first exceed the threshold. Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β .

A central implication of this framework is that predictions depend strongly on the underlying temperature regime during which the plant accumulates temperature. When plants accumulate temperatures during times when temperature is relatively flat (stationary, common in winter months following the winter solstice) cumulative temperature grows at an approximately linear rate; thus the expected bloom date advances with climate change in step with increasing baseline temperatures ($\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}$) as is often expected in the literature (vitasse2022great). In contrast, when plants may accumulate temperatures mainly during the spring months when average daily temperatures increase approximately linearly over time and thus cumulative temperatures grow quadratically. In this regime, bloom date advances as a function of the rate of within-season warming ($\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}$)—that is, how quickly spring temperatures increase.

In both regimes, increasing temperatures advance bloom dates in a non-linear function, similar to shifts in a suite of recent papers (fu2015declining, wolkovich2021simple). Based on either regime, increasing temperatures advance bloom dates with diminishing advances as the rate of warming (to the baseline temperature, α or within-season rate, β) increases relative to the size of the total threshold (for the stationary regime: $\frac{\partial}{\partial \alpha} \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}$; for the within-season increasing temperatures regime: $\frac{\partial}{\partial \beta} \mathbb{E}[\nu(t)] \approx -\frac{1}{2} \sqrt{\frac{2t}{\beta^3}}$). If climate change shifts the period when plants accumulate temperatures from the non-stationary springtime period (near the spring equinox) to the more stationary period earlier in the year (near the winter solstice) then advances may actually increase, particularly if increases in the baseline (α) outpace increases in the within-season rate (β).

Importantly, these two temperature regimes make very different predictions for the synchrony of events—that is the variance in event timings within the same year and location. Both predict decreased variance with warming (stationary regime: $\text{Var}(\nu(t)) \approx \frac{\sigma^2 t}{\alpha^3}$; non-stationary regime: $\text{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2} \sqrt{2t}}$), but the influence of noise is strikingly different. In the non-stationary regime—common in the spring—noise is quickly overwhelmed by the effect of increasing within-season temperatures. This effectively acts to synchronize plants (and buds within plants) with the same thermal threshold (aka GDD) requirement—making small differences in microclimate or other idiosyncratic effects negligible. In contrast, under the stationary regime—common in winter—noise can remain relatively influential so that small microclimate differences translate into larger differences in threshold-crossing times. This prediction fits well with experiments conducted under constant temperature regimes, which find increasing variance with decreasing baseline temperatures (Charrier:2011aa). In such experiments, buds often have extremely variable budburst timings, exactly as our model predicts (see

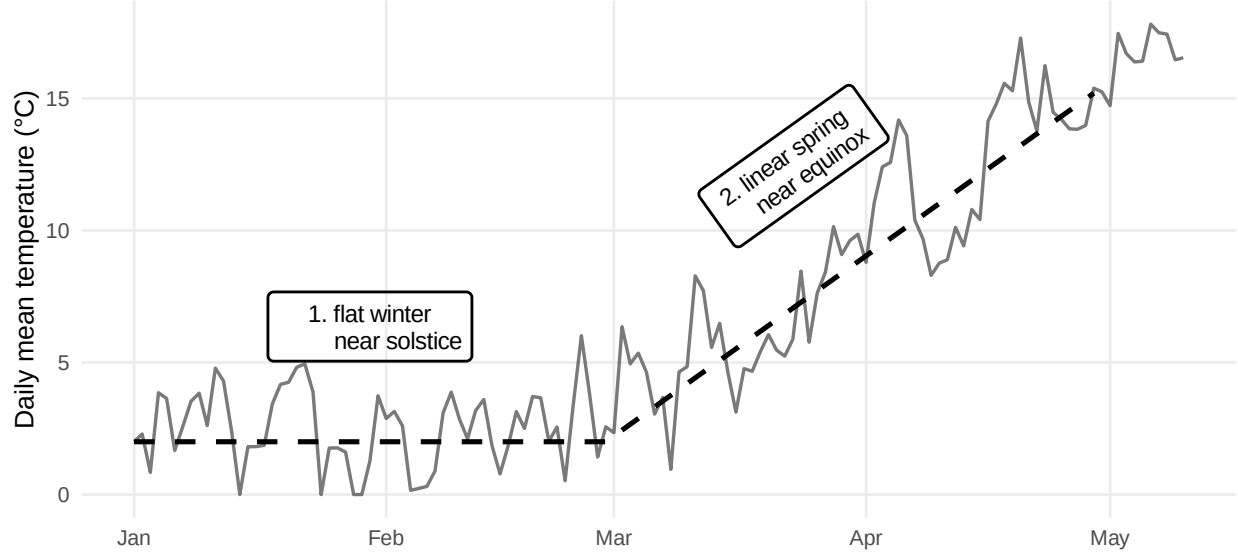


Figure 1: The timing of springtime events in temperate climates depends on whether temperature is accumulated during one of two regimes, 1. a flat winter or 2. a linear spring.

Appendix).

These contrasting predictions for variance based on the underlying temperature regime predict increasing variance when anthropogenic warming produces regime shifts. As warming advances events earlier in the calendar, thresholds are reached in portions of the season with flatter mean temperatures where timing can be more sensitive to local variation. In such cases, warming produces a regime shift that simultaneously (i) advances springtime events potentially at a faster rate and (ii) reduces synchrony across nearby locations, producing a more variable and less coordinated onset of spring even as it arrives sooner. This regime shift produces a “scattered spring” when climate change moves events earlier on average while simultaneously reducing coordination across individuals and locations, making spring less tightly synchronized even as it arrives sooner.

Whether warming leads to a more synchronized or more “scattered” spring therefore depends on the local joint evolution of (α, β) and on the threshold t relevant for a given species and life-cycle event. In this sense, the stopped-random-walk framework offers a simple way to translate spatially varying climate change into spatially varying predictions about both the advancement of spring and the stability of ecosystem coordination. We examine this using data from the historical lilac-honeysuckle phenology network compiled by Rosemartin et al. (2015), which contains leafing and flowering dates collected across the continental United States from 1956–2014 for purple common lilac (*Syringa vulgaris*), a cloned lilac cultivar (*S.*, $\times$, *chinensis* ‘Red Rothomagensis’), and two cloned honeysuckle cultivars (*Lonicera tatarica* Arnold Red and *L. korolkowii* Zabeli).

The exact effect on an ecosystem depends on how exactly the baseline (α) and increasing within season (β)

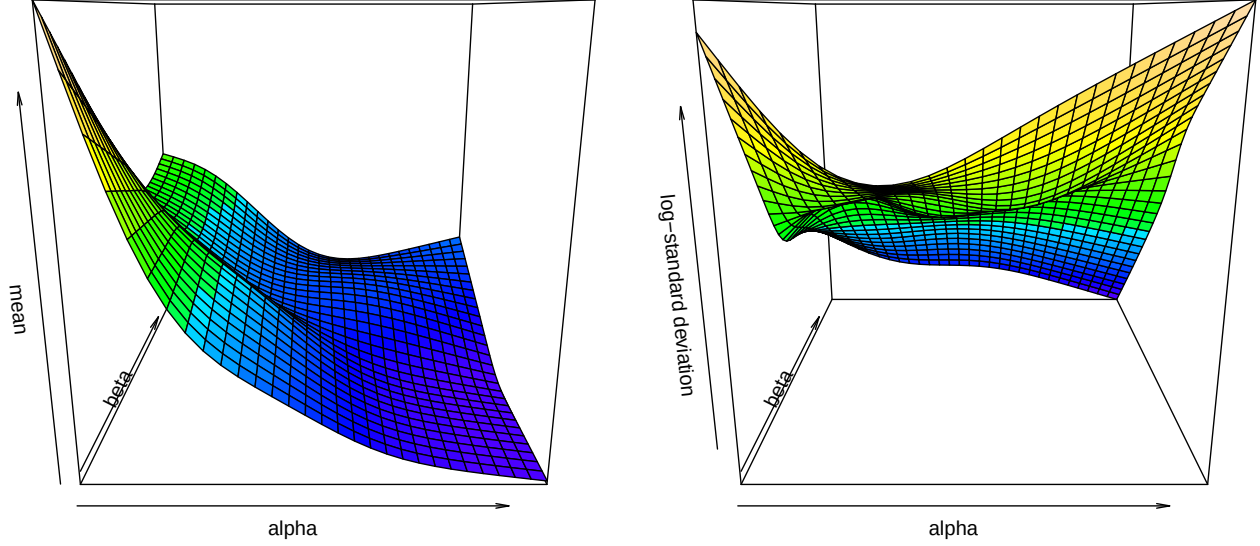


Figure 2: Mean and log standard deviation of lilac bloom dates ($n = 23,864$) by baseline temperatures (α) and within-season temperature increases (β) estimated by a generalized additive model. The left figure shows the mean generally decreases at an increasing rate as α and β increase—although increases are possible when β is large. The standard deviation can increase or decrease, again depending on the relative values of α or β .

temperatures change. In some locations, climate change may manifest primarily as higher winter temperatures (an increase in α) with relatively little change in the rate at which temperatures rise during spring (a modest change in β). In others, winter temperatures may change less while spring transitions steepen (a larger increase in β). Because the mean and variance of the hitting time depend differently on α and β , these distinct warming profiles can produce different phenological responses even when average annual warming is similar.

To test if baseline winter and increasing spring temperatures shift flowering times as predicted by our model, we first bin sites and years within a grid of α (rows) and β (columns) and calculate the mean and standard deviation of the observed bloom date (number of days since January 1) within each bin. We also checked these results by replacing the binning with a smooth, nonparametric estimate of the conditional mean and dispersion using generalized additive models (GAM), see Hastie and Tibshirani (1986) and Wood (2017).

Across both approaches (binning and GAM) shifts in bloom dates across varying baseline winter and increasing spring temperatures generally supported our model predictions. First, holding α fixed, larger β is associated with earlier mean bloom dates, which generally but not always advance at a decreasing rate. The same is true for holding β fixed, and increasing α . Holding α at its largest value, the influence of β is considerably smaller. This reflects the accuracy of predictions when plants accumulate all temperatures under a stationary climate where increasing temperatures (β) plays no role.

Consistent with the theory, dispersion generally declined with higher increasing spring temperatures (i.e., as β increases within each α row). This occurs because when baseline temperatures (α) are held fixed, the percent

Table 1: Mean bloom date (day-of-year) by average Jan-Feb temperature α (rows) and average Mar-Apr temperature increase β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
$(0, 0.6]$	163.05	154.24	148.96	142.56
$(0.6, 1.6]$	153.21	147.97	142.60	136.78
$(1.6, 4]$	138.86	133.56	130.06	126.34
$(4, 16.4]$	105.13	106.46	105.26	102.67

of buds that bloom in the stationary climate is unchanged, and the buds that makes it to the increasing temperature regime have a lower variance. However, holding β fixed and increasing α produces a regime shift in which more buds bloom under the stationary climate regime and thus the bloom date dispersion increases. This is the “scattered spring” predicted from our model. Warmer winters advance bloom dates in to an earlier portion of the seasonal cycle in which forcing is effectively flatter and cumulative temperatures grow more slowly. We find similar results in the GAM (Figure 2, which show the mean (left) and log standard deviation (right) plotted again alpha and beta).

Our results do not exclude the possibility of chilling ...

Methods

We focus site that recorded the phenophase full bloom and are within 10 miles of a GHCND site with complete temperature records, resulting in 23,864 observations.

For each observation, we estimate α using the average daily temperature between January and February. We estimate β using the slope of an linear regression model fit by regressing daily temperature on day index over March and April. Temperatures below 0 are set to 0. The parameters α and β are shown for nine random sites in Figures 4 and 5 in Appendix A.2, which have the same interpretation as Figure 1. Site 14726 in New Mexico (Year 1969) is a good example of a bloom in regime 2 since relatively little temperature was accumulated during January and February. The average bloom date was May 8th with a standard deviation of 17 days. Site 14276 in New York (Year 1982) is a good example of a bloom between regimes 1 and 2, which had an average bloom date of March 18 with a standard deviation of 27.

We fit a two-parameter Gaussian location–scale model using the `gaulss` function from `mgcv`:

$$\mu(\alpha, \beta) = \mathbb{E}[\nu(\tau) \mid \alpha, \beta], \quad \sigma(\alpha, \beta) = \text{SD}(\nu(\tau) \mid \alpha, \beta),$$

where both μ and σ are modeled with tensor-product splines with basis dimension $k = 10$. This approach uses all observations while allowing flexible interactions between the two temperature summaries.

Table 2: Standard deviation of bloom date (days) by winter level α (rows) and spring warming rate β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
$(0, 0.6]$	16.08	12.36	10.44	9.61
$(0.6, 1.6]$	15.24	12.87	13.20	12.67
$(1.6, 4]$	17.97	16.52	16.31	14.56
$(4, 16.4]$	22.78	19.84	17.23	14.76

- Charrier, Guillaume, Marc Bonhomme, André Lacointe, and Thierry Améglio. 2011. “Are Budburst Dates, Dormancy and Cold Acclimation in Walnut Trees (*Juglans Regia* l.) Under Mainly Genotypic or Environmental Control?” *International Journal of Biometeorology* 55 (6): 763–74.
- Gut, Allan. 2009. *Stopped Random Walks*. Springer. <https://isbnsearch.org/isbn/9780387878348>.
- . 2011. “Renewal Theory with a Trend.” *Statistics & Probability Letters* 81 (8): 1292–99. <https://doi.org/10.1016/j.spl.2011.03.031>.
- . 2012. “Anscombe’s Theorem 60 Years Later.” *Sequential Analysis* 31 (3): 368–96. <https://www.tandfonline.com/doi/10.1080/07474946.2012.694349>.
- Hacking, Ian. 1990. *The Taming of Chance*. 17. Cambridge University Press.
- Hastie, Trevor, and Robert Tibshirani. 1986. “Generalized Additive Models.” *Statistical Science* 1 (3): 297–310.
- Lai, Tze L, and David Siegmund. 1977. “A nonlinear renewal theory with applications to sequential analysis I.” *The Annals of Statistics*, 946–54.
- Pearson, Karl. 1905. “The Problem of the Random Walk.” *Nature* 72 (1867): 342–42.
- Quetelet, Adolphe. 1849. *Letters Addressed to HRH the Grand Duke of Saxe Coburg and Gotha: On the Theory of Probabilities, as Applied to the Moral and Political Sciences*. C. & E. Layton.
- Rosemartin, Alyssa H, Ellen G Denny, Jake F Weltzin, R Lee Marsh, Bruce E Wilson, Hamed Mehdipoor, Raul Zurita-Milla, and Mark D Schwartz. 2015. “Lilac and Honeysuckle Phenology Data 1956–2014.” *Scientific Data* 2 (1): 1–8.
- Wald, Abraham. 1992. “Sequential Tests of Statistical Hypotheses.” In *Breakthroughs in Statistics: Foundations and Basic Theory*, 256–98. Springer.
- Wood, Simon N. 2017. *Generalized Additive Models: An Introduction with r*. chapman; hall/CRC.
- Woodroffe, Michael. 1982. *Nonlinear Renewal Theory in Sequential Analysis*. SIAM.
- Yongshuo H Fu, Hongfang Zhao, Shilong Piao, Marc Peaucelle, Shushi Peng, Guiyun Zhou, Philippe Ciais, Mengtian Huang, Annette Menzel, Josep Peñuelas, et al. Declining global warming effects on the phenology of spring leaf unfolding. *Nature*, 526(7571):104–107, 2015.
- Yann Vitasse, Constant Signarbieux, and Yongshuo H Fu. Global warming leads to more uniform spring phenology across elevations. *Proceedings of the National Academy of Sciences*, 115(5):1004–1008, 2018.
- Michael Stemkovski, James R Bell, Elizabeth R Ellwood, Brian D Inouye, Hiromi Kobori, Sang Don Lee,

- Trevor Lloyd-Evans, Richard B Primack, Barbara Templ, and William D Pearse. Disorder or a new order: how climate change affects phenological variability. *Ecology*, 104(1):e3846, 2023.
- Isabelle Chuine, Iñaki Garcia de Cortazar-Atauri, Frédéric Jean, and Colin Van Reeth. Living things are showing increasing anomalies in their seasonal activity, which could disrupt the dynamics of biodiversity and ecosystems. *Scientific Reports*, 15(1):32860, 2025.
- EM Wolkovich, J Auerbach, CJ Chamberlain, DM Buonaiuto, AK Ettinger, I Morales-Castilla, and A Gelman. A simple explanation for declining temperature sensitivity with warming. *Global Change Biology*, 27(20):4947–4949, 2021.
- Xiaojie Gao, Andrew D Richardson, Mark A Friedl, Minkyu Moon, and Josh M Gray. Thermal forcing versus chilling? misspecification of temperature controls in spring phenology models. *Global Ecology and Biogeography*, 33(12):e13932, 2024.
- Mark D Schwartz, Jonathan M Hanes, and Liang Liang. Separating temperature from other factors in phenological measurements. *International journal of biometeorology*, 58(7):1699–1704, 2014.
- Marc Peaucelle, Josep Penuelas, and Hans Verbeeck. Accurate phenology analyses require bud traits and energy budgets. *Nature Plants*, 8(8):915–922, 2022.

Summary of analytical solution

To fix ideas, consider the day a bud blooms at a given location. We model the effective temperature experienced by that bud on day i by a linear trend plus microclimate noise,

$$X_i = \alpha + \beta i + \epsilon_i, \quad (1)$$

where α is a baseline level, β is a within-season warming rate, and $\{\epsilon_i\}_{i \geq 1}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$, representing idiosyncratic variation at that bud such as due to the day-to-day microclimate.

Let

$$Z_n = \sum_{i=1}^n X_i \quad \text{and} \quad \nu(t) = \min\{m : Z_m > t\} \quad (2)$$

denote cumulative temperature by day n and the observed bloom date, respectively, where $t > 0$ is a plant-specific temperature-sum threshold. Thus $\nu(t)$ is the hitting time at which cumulative temperatures first exceed the threshold.

Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β . We consider two cases.

Regime 1: constant drift ($\beta = 0$) When the drift is approximately constant and positive,

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right), \quad (3)$$

see Gut (2011) and Gut (2009, chap. 3).

Regime 2: increasing drift ($\beta > 0$) When average temperatures increase, the hitting time satisfies

$$\nu(t) \sim \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2}, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right), \quad (4)$$

with the derivation sketched in Appendix A.1 and simulations confirming the derivations in Appendix A.3. The functional central limit theorem and linearization argument underlying these results hold somewhat generally, and thus similar limits can be derived when the $\{\epsilon_i\}_{i \geq 1}$ are weakly correlated or the temperature trajectory is not linear but sufficiently.

From equations (3)–(4), we make two observations. Finding 1: there is a nonlinear relationship between warming and the mean timing of springtime events, and warming can advance springtime events at an increasing rate. Finding 2: the two regimes are not equally sensitive to the idiosyncratic variation, and thus warming can increase the variation of springtime events, thereby scattering spring. We now discuss both

findings in greater detail.

2.1 Finding 1: warming can advance springtime events at an increasing rate

In both regimes, the expected bloom date is a nonlinear function of the warming parameters.

Regime 1: constant drift ($\beta = 0$) From (3),

$$\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}, \quad \text{Var}(\nu(t)) \approx \frac{\sigma^2 t}{\alpha^3}. \quad (5)$$

Increasing baseline temperatures (larger α) advances the expected bloom date, but the marginal advance diminishes since

$$\frac{\partial}{\partial \alpha} \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}, \quad (6)$$

where the magnitude decreases as α grows.

Regime 2: increasing drift ($\beta > 0$) From (4), when t is large,

$$\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}, \quad \text{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2} \sqrt{2t}}. \quad (7)$$

Increasing the within-regime warming rate (larger β) also advances the expected bloom date with diminishing returns since

$$\frac{\partial}{\partial \beta} \mathbb{E}[\nu(t)] \approx -\frac{1}{2} \sqrt{\frac{2t}{\beta^3}}. \quad (8)$$

However, if warming climates shift the bloom date from regime 2 to regime 1, then the rate of advancement can increase, particularly if increases in α outpace increases in β .

2.2 Finding 2: warming can scatter spring events

Equations (3)–(4) also clarify how warming affects dispersion. Holding the regime fixed, warming reduces variance. That is, $\text{Var}(\nu(t))$ decreases with α in (3) and with β in (4). However, just as in Section 2.1, warming can change which regime is relevant at the realized bloom date. When an event is advanced sufficiently early, microclimate fluctuations play a larger role in determining the hitting time.

One way to quantify sensitivity to microclimate variation is through the (squared) coefficient of variation,

$$\text{CV}^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2}.$$

In regime 1, $\mathbb{E}[\nu(t)] \approx t/\alpha$ and $\text{Var}(\nu(t)) \approx \sigma^2 t/\alpha^3$, so

$$\text{CV}_1^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \approx \frac{\sigma^2 t/\alpha^3}{(t/\alpha)^2} = \frac{\sigma^2}{\alpha t}.$$

In regime 2, $\mathbb{E}[\nu(t)] \approx \sqrt{2t/\beta}$ and $\text{Var}(\nu(t)) \approx \sigma^2/(\beta^{3/2}\sqrt{2t})$, hence

$$\text{CV}_2^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \approx \frac{\sigma^2/(\beta^{3/2}\sqrt{2t})}{(2t/\beta)} = \frac{\sigma^2}{\sqrt{\beta}(2t)^{3/2}}.$$

Both expressions increase when deterministic accumulation is weaker relative to the microclimate scale σ (e.g., smaller α and/or smaller effective β). Consequently, warming tends to reduce variability within a given seasonal regime by strengthening deterministic forcing.

To compare regimes, note that

$$\frac{\text{CV}_1^2}{\text{CV}_2^2} \approx \frac{\sigma^2/(\alpha t)}{\sigma^2/(\sqrt{\beta}(2t)^{3/2})} = \frac{\sqrt{\beta}2^{3/2}}{\alpha} \sqrt{t}.$$

Thus, since α and β are nearly always greater than 0, the regime 1 becomes arbitrarily more sensitive to microclimate fluctuations than regime 2 for large t . This produces a more variable, less synchronized and thus “scattered” spring when warming shifts threshold crossings from regime 2 to regime 1.

Experimental Data

We first consider a controlled experiment on walnut trees (*Juglans*-sp.) conducted by Charrier et al. (2011). The data examines 30 trees comprising 6 genotypes at 2 locations. In November of the study year, stems were sampled from each tree and cut into 7 centimeter segments containing a single bud. All segments were chilled at 4°C and then transferred to constant “forcing” environments with temperatures set at one of 5, 10, 15, 20, or 25°C. For each temperature, the outcome is the response time, the time (in days) until budburst determined from the date at which 50% of buds reached stage 15 of the BBCH scale.

This experiment isolates the winter regime because temperature is held constant over time. Recall that in such cases, cumulative forcing grows approximately linearly,

$$Z_n \approx \alpha n + \sum_{i=1}^n e_i,$$

so that the hitting time $\nu(t)$ satisfies the approximation

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right),$$

implying that increasing α should (i) reduce the mean time to budburst and (ii) reduce dispersion. In addition, the squared coefficient of variation,

$$\text{CV}^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2},$$

provides a unit-free measure of sensitivity to stochastic fluctuations, which should decrease as α increases.

Table 3: Summary data from the Walnut forcing experiment [charrier2011budburst].

α	n	mean	sd	cv ²
5	30	178	27.28	0.0236
10	30	88	12.44	0.0199
15	30	39	11.33	0.0831
20	30	26	7.67	0.0851
25	30	20	4.45	0.0480

Table 3 reports the mean and standard deviation of the response times across the 30 stems at each forcing temperature, along with the sensitivity measure CV². The pattern is consistent with the regime 1 model. Higher forcing temperatures substantially advance budburst (the mean falls sharply with α), and the dispersion of budburst timing is smaller at warmer forcing temperatures. Thus, in an experimental setting, the stopped-random-walk approximation captures the basic comparative statics of constant-temperature forcing on both the level and variability of budburst times. (The sensitivity does decline from $\alpha = 20$ to $\alpha = 25$. The reason for low sensitivity for $\alpha = 5$ and 10 is unclear and may reflect censoring.)

A. Appendix

A.1 Sketch of Theoretical Results

We follow the notation in Gut (2009, chap. 6). Let $X_i = \alpha + \beta i + \epsilon_i$ denote the temperature on day i , where $\alpha > 0$, $\beta \geq 0$, $\mathbb{E}[\epsilon_i] = 0$, and $\text{Var}(\epsilon_i) = \sigma^2$. Define the deterministic cumulative sum

$$\xi_n = \sum_{i=1}^n \mathbb{E}[X_i] = \alpha n + \frac{\beta}{2}n(n+1) = \frac{\beta}{2}n^2 + \beta\gamma n, \quad \gamma = \frac{\alpha}{\beta} + \frac{1}{2},$$

with $\xi_n = \alpha n$ when $\beta = 0$. Denote the sum of the stochastic component $S_n = \sum_{i=1}^n \epsilon_i$. Our object of interest is the first-passage time of

$$Z_n = S_n + \xi_n, \quad \nu(t) = \min\{n \geq 1 : Z_n > t\}.$$

In the language of Lai and Siegmund (1977), $\{Z_n\}$ is a perturbed random walk: the random walk $\{S_n\}$ is perturbed by a deterministic term $\{\xi_n\}$, and when $\beta > 0$ the quadratic growth of ξ_n dominates long-run behavior. This setup yields two asymptotic regimes, corresponding to the empirical distinction between regime 1 ($\beta \approx 0$) and regime 2 ($\beta > 0$), which in turn underlies our qualitative predictions about nonlinear mean shifts and changes in dispersion.

In regime 1, $\beta = 0$, and we apply the usual asymptotic argument for the hitting time of a random walk with constant positive drift.

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right),$$

see Gut (2011) and Gut (2009, chap. 3).

In regime 2, $\beta > 0$, and the deterministic component $\xi_n = \frac{\beta}{2}n^2 + \beta\gamma n$ grows quadratically so that the threshold t is reached near the deterministic crossing time $m(t)$ solving $\xi_{m(t)} \approx t$, i.e.

$$m(t) = \frac{-\beta\gamma + \sqrt{\beta^2\gamma^2 + 2\beta t}}{\beta} = \sqrt{\frac{2t}{\beta}} - \gamma + o(1).$$

By the functional central limit theorem, the deviation of $\nu(t)$ about $m(t)$ is approximately normal. Linearization of $Z_{\nu(t)} \approx t$ around $m(t)$ yields

$$\nu(t) \sim \text{Normal}\left(m(t), \frac{\sigma^2 m(t)}{(\alpha + \beta m(t))^2}\right) = \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \gamma, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right),$$

where we have used the fact that $\alpha + \beta m(t) \sim \beta m(t) \sim \sqrt{2\beta t}$ for $t \gg 0$. Thus, in regime 2 the mean hitting time scales like $\sqrt{t}$ (rather than linearly in t) and the variance shrinks at rate $t^{-1/2}$, reflecting that increasing forcing makes microclimate fluctuations progressively less influential near the crossing.

A.2 Additional Figures

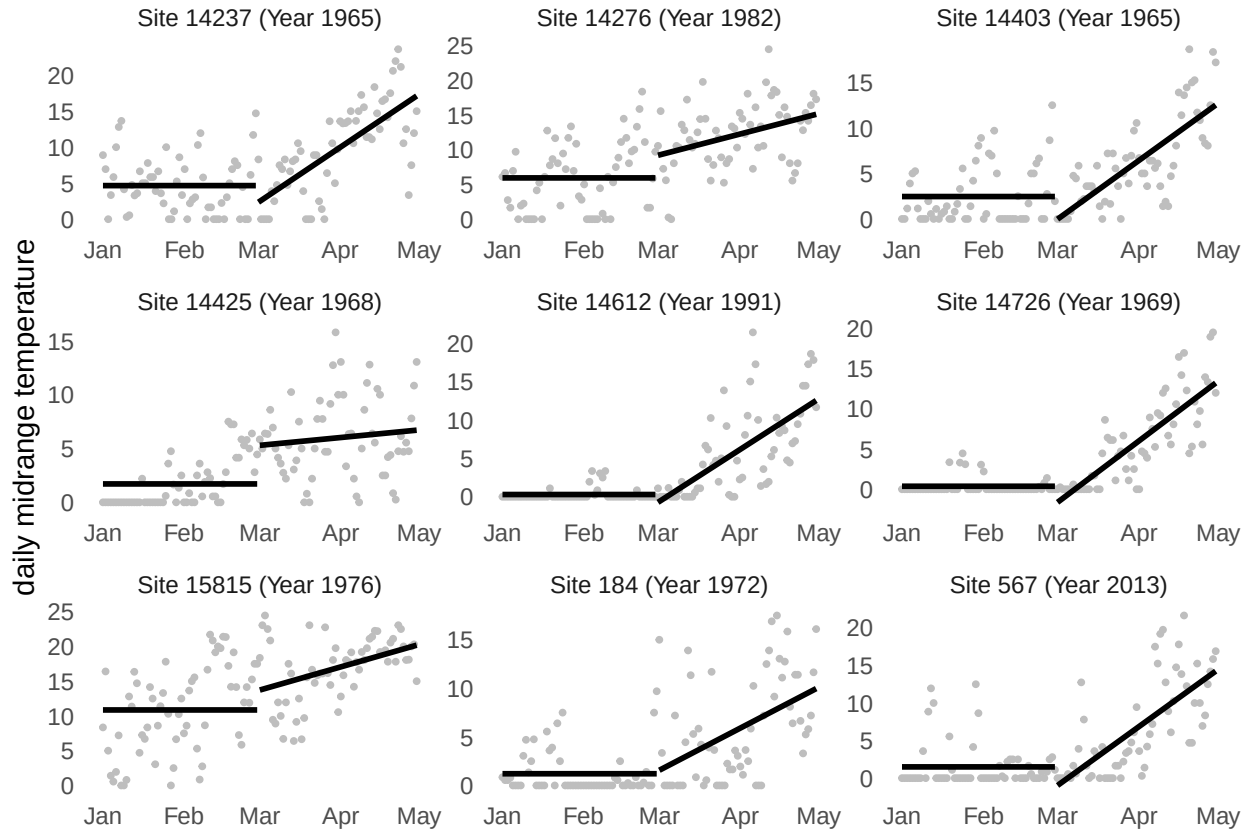


Figure 3: Springtime daily midrange temperatures (points) with α (mean from Jan to Feb) and β (slope from Mar to Apr) for nine randomly selected sites. Temperatures below 0 are set to 0.

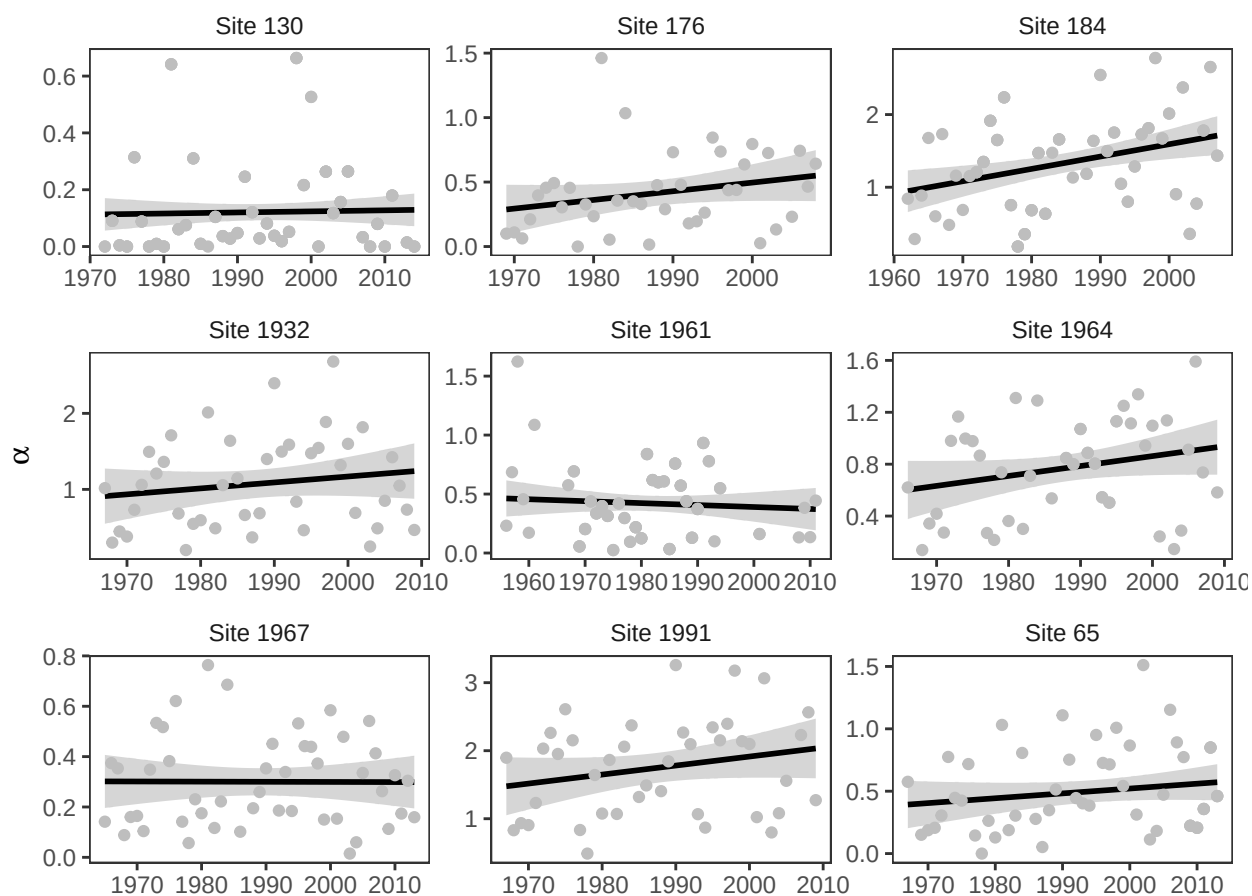


Figure 4: Changes in alpha for 9 sites in lilac data.

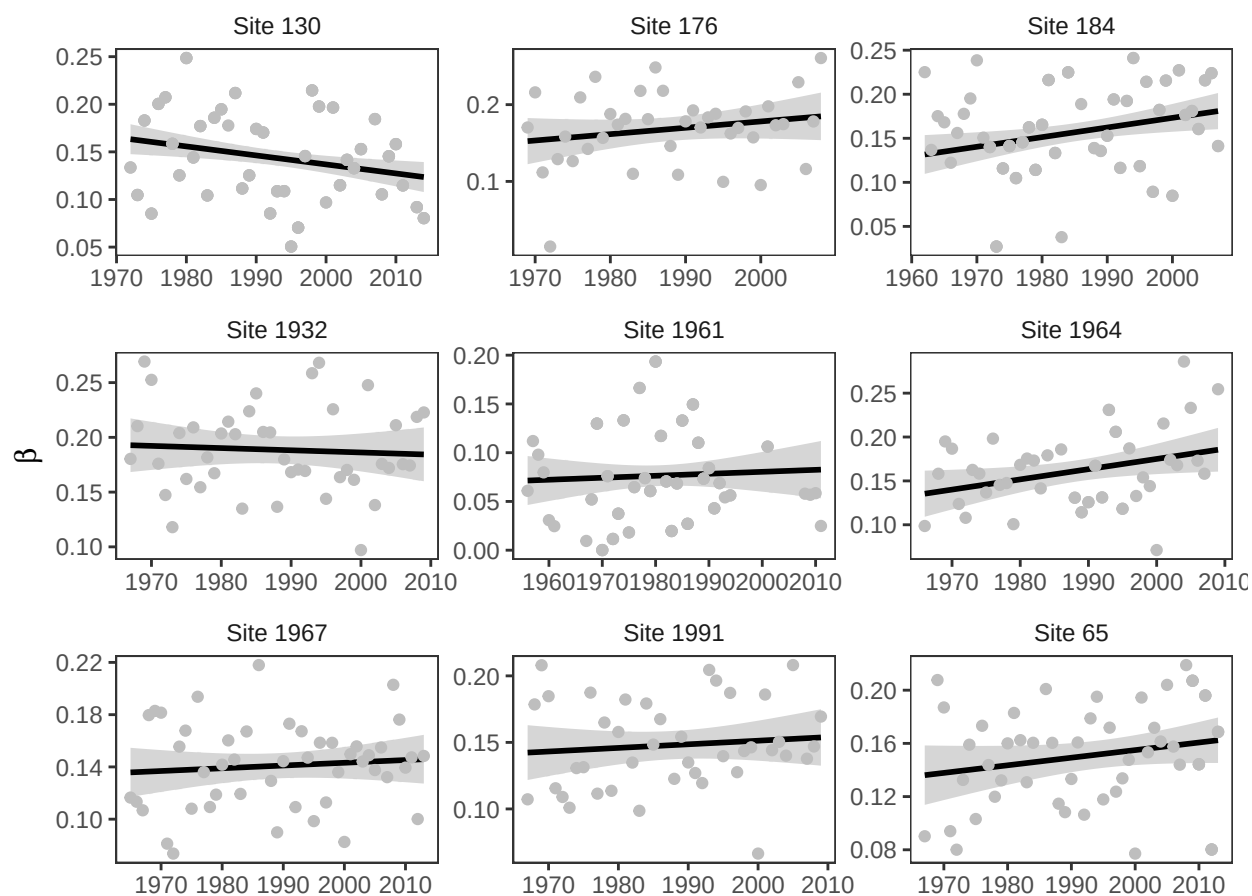


Figure 5: Changes in beta for 9 sites in lilac data.

A.3 Simulations

We run two simulations to study the results from Sections 2 and 3. The first checks the asymptotic approximation from Section 2. The second checks the interpretation of the data from Section 3.

Simulation 1

The first simulation verifies the accuracy of the two asymptotic normal approximations for the hitting time

$$\nu(t) = \min \left\{ n \geq 1 : \sum_{i=1}^n X_i > t \right\}$$

under the model

$$X_i = \alpha + \beta i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma^2).$$

For simulation, we set $\sigma = 20$ and generate daily temperatures until the threshold t is reached, recording the corresponding hitting time $\nu(t)$. We repeat this procedure $R = 10,000$ times for thresholds $t = \{1000, 2000\}$, $\alpha = \{2, 4\}$ and $\beta = \{0, 0.1\}$. When $\beta = 0$ the simulation matches the winter regimes, and when $\beta = 0.1$, it matches the spring regime. The simulated distribution of $\nu(t)$ is represented in Figure 6 below by the histograms.

We then compare these simulations with the theoretical approximations, we standardized each simulated hitting time using the relevant asymptotic mean and standard deviation. When $\beta = 0$, we use the winter regime approximation

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right).$$

When $\beta > 0$, we use the spring regime approximation

$$\nu(t) \sim \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2}, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right).$$

For each setting we computed the standardized variable

$$z = \frac{\nu(t) - \mu}{\text{sd}},$$

plotted its histogram, and overlaid the standard normal density. Across both regimes, the standardized histograms align closely with the $\text{Normal}(0, 1)$ curve, and the agreement improves as t increases, providing a visual confirmation of the derived asymptotic distributions.

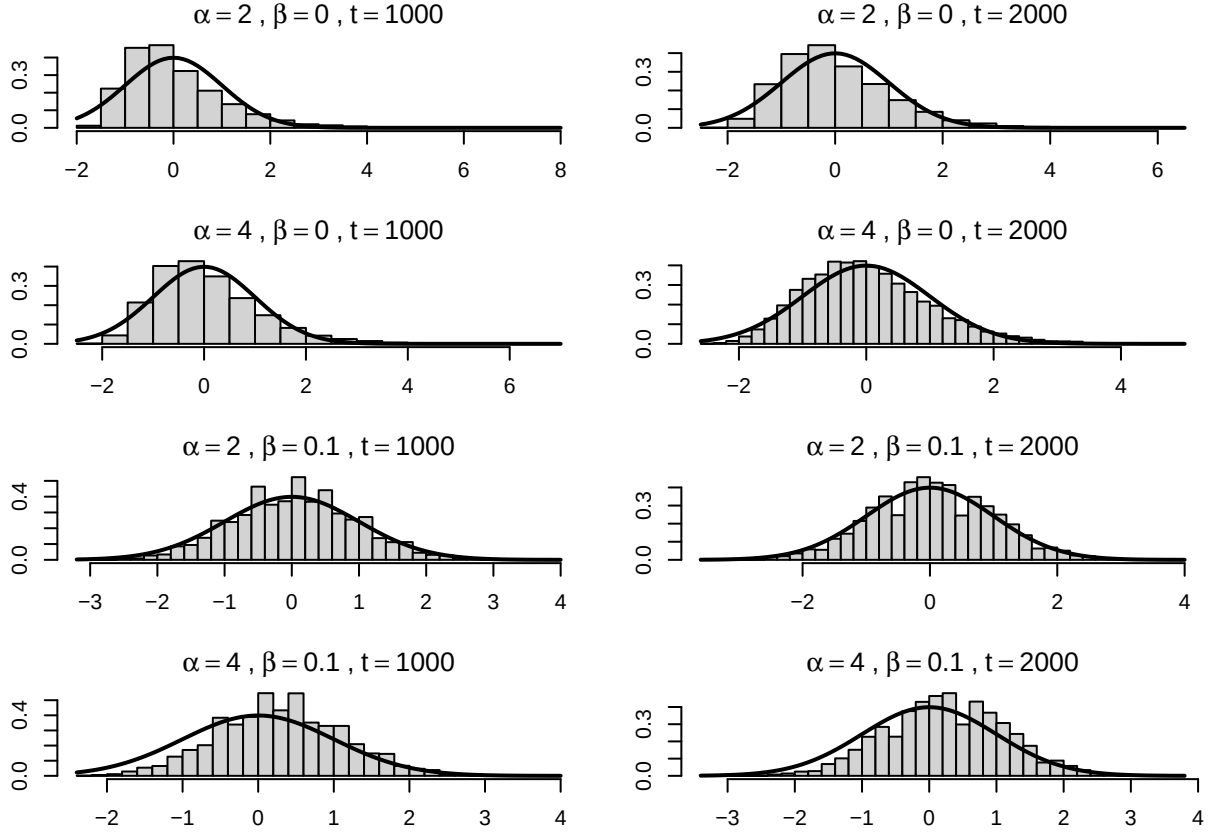


Figure 6: Simulations comparing the distribution of bloom dates under the winter (top) and spring (bottom) regimes.

Simulation 2

To verify our interpretation of the data in Section 3, we simulate the model from section. In each replicate we generated daily temperatures

$$X_i = \mu_i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma^2),$$

with $\sigma = 20$, where the deterministic component follows

$$\mu_i = \begin{cases} \alpha, & i = 1, \dots, 90 \quad (\text{January–February}), \\ \alpha + \beta(i - 90), & i = 91, \dots, 180 \quad (\text{March–April}). \end{cases}$$

Bloom occurs when cumulative temperature first exceeds a threshold t , i.e.

$$\nu(t) = \min \left\{ n \geq 1 : \sum_{i=1}^n X_i > t \right\}.$$

We simulated $R = 10,000$ independent realizations of $\nu(t)$ for each combination of $\alpha \in \{4, 8, 10\}$ and $\beta \in \{0.2, 0.4, 0.8\}$, and for thresholds $t \in \{1000, 2000\}$. For each parameter pair we report the Monte Carlo mean and standard deviation of $\nu(t)$ (Tables~4a–4d), arranged with rows indexing α and columns indexing β .

The results reproduce the qualitative patterns observed in the lilac application. Mean bloom dates decrease as either the winter level α increases or the spring warming rate β increases (Tables~4a and 4c). Within a fixed α row, increasing β reduces the standard deviation of bloom dates (Tables~4b and 4d), consistent with the spring-regime prediction that stronger increasing forcing diminishes the leverage of noise on the threshold-crossing time. Holding β fixed, dispersion is often larger at higher α (particularly for $\beta \in \{0.4, 0.8\}$), consistent with the regime-shift intuition that earlier crossings can occur in a flatter portion of the seasonal cycle where microclimate variability has greater influence.

Table 4: Piecewise winter–spring simulations. Simulated mean and standard deviation of the bloom date $\nu(t)$ across $R = 10,000$ Monte Carlo replicates, for thresholds $t \in \{1000, 2000\}$. Rows index the winter level α and columns index the spring warming rate β .

(a) Mean, $t = 1000$				(c) Mean, $t = 2000$			
$\alpha \setminus \beta$	0.2	0.4	0.8	$\alpha \setminus \beta$	0.2	0.4	0.8
4	151.66	136.83	124.86	4	199.54	171.15	149.16
8	114.87	111.27	106.85	8	169.81	152.43	137.37
10	98.45	97.20	95.72	10	156.22	143.40	131.34

(b) SD, $t = 1000$				(d) SD, $t = 2000$			
$\alpha \setminus \beta$	0.2	0.4	0.8	$\alpha \setminus \beta$	0.2	0.4	0.8
4	15.48	10.42	7.21	4	10.96	7.22	4.84
8	16.53	13.27	10.50	8	10.93	7.50	5.11
10	15.94	14.45	12.71	10	10.69	7.66	5.36